CHEMISTRY PAPER 1

SECTION A (54 MARKS)

1. Standard electrode potentials for some half cell reactions are given below.

				E^{θ} (v)
R-	Fe^{3+} (aq) + e	$\longrightarrow$	Fe^{2+} (aq)	+0.77
S-	Cl_2 (g) + 2e	$\longrightarrow$	$2Cl^-$ (aq)	+1.76

a) Write the cell notation for the cell formed when the two half cells are combined. (1 mark)

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b) Write half equation for the reaction that takes place at the ;

i) Anode (1 mark)

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ii) Cathode (1 mark)

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c) Write the equation for overall cell reaction. (1 ½ marks)

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d) Calculate Gibb's free energy (ΔG^{θ}) for the reaction and state its units.

(Farady constant = 95650C) (1 ½ marks)

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2. a) (i) Write the electronic configuration of silicon. (0 ½ mark)

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(ii) State the common oxidation state(s) exhibited by silicon in its compounds.

(1 mark)

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b) Write equations for the reaction between silicon and;

i) water (1 ½ marks)

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ii) magnesium (1 ½ marks)

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iii) hydrofluoric acid (1 ½ marks)

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3. Write equations to show how the following conversions can be effected.

a) CH_3CH_2Cl to $(CH_3CH_2)_2O$ (2 marks)

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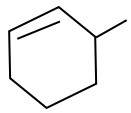
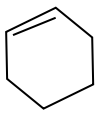
b) 



to (3 marks)

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4. a) Write equations for hydrolysis of chromium(III) sulphate. (1 ½ marks)

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b) Magnesium ribbon was added to the resultant solution in (a).

i) state what was observed. (0 ½ marks)

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ii) Write equation for the reaction that takes place. (0 ½ marks)

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5. a) Define the term buffer solution. (1 mark)

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b) Calculate the pH of a mixture containing $20cm^3$ of 0.1M sodium hydroxide and $100cm^3$ 0f 0.1M ethanoic acid. (Ka for ethanoic acid at 25^0C = $1.80x10^{-5}moldm^{-3}$)

(3 marks)

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6. a) What is meant by the term bond energy? (1 mark)

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b) Bond energies of halogens are shown in the table below.

Element	Fluorine	Chlorine	Bromine	Iodine
Bond energy	+158	+242	+193	+151

i) State how bond energies vary. (0 ½ mark)

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ii) Explain your answer in (i) (2 ½ marks)

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7. $10cm^3$ of a gaseous hydrocarbon (Z), (C_xH_y) was ignited in $60cm^3$ of oxygen in excess. On cooling to room temperature, the residual gas occupied $50cm^3$.

When the residual gas was treated with concentrated sodium hydroxide solution, the volume reduced by $20cm^3$.

a) (i) Write the equation for the reaction between Z and Oxygen. (1 ½ marks)

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(ii) Calculate the molecular formula of Z. (2 ½ marks)

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b) Write equations to show how Z can be prepared from calcium oxide.

(2 marks)

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8. Spathic iron , Haematite and magnetite are some of the ores used to extract pig iron.

   a) Write the chemical formulae of

      i) Spathic iron (0 ½ mark)

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      ii) Haematite (0 ½ mark)

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      iii) Magnetite (0 ½ mark)

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   b) Name two physical methods used to convert spathic iron ore to pig iron.

      (1 mark)

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   c) Write three principal reactions that occur in the blast furnace leading to formation of pig iron. (3 marks)

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9. a) Draw the structure and name the shape of the species in the table below.

(3 marks)

Species	Structure	Shape
IO_3^-		
NO_2^-		

b) Write equation for the reaction between

i) Aqueous potassium Iodate(V) solution and acidified potassium iodide.

(1 ½ marks)

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ii) Aqueous sodium nitrite solution and acidified potassium dichromate (VI) solution. (1 ½ marks)

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SECTION B (54 marks)

10. a) Write equation(s) for the reaction between chlorine and;

i) Warm concentrated potassium hydroxide solution. (1 ½ marks)

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ii) Cold dilute potassium hydroxide solution. (1 ½ marks)

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b) (i) State condition(s) under which chlorine reacts with sulphur dioxide.

(1 mark)

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(ii) Write equation for the reaction. (1 mark)

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c) Chlorine was bubbled through sodium thiosulphate and to the resultant solution was added aqueous lead (II) nitrate solution.

i) State what was observed. (1 mark)

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ii) Write equation(s) for the reaction(s) that take place. (2 marks)

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11. When 6.8g of a compound W containing carbon, hydrogen and oxygen was burnt in excess air, 17.6g of carbon dioxide and 3.6g of water was formed.

a) Determine the empirical formula of W. (3 marks)

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b) Calculate the molecular formula of W(Vapour density of W is 68)

(1 ½ marks)

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c) W burns with a sooty flame and reacts with aqueous sodium bicarbonate solution evolving carbon dioxide.

i) Identify W (0 ½ mark)

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ii) A Mixture of W and ethanol was heated in the presence of

concentrated sulphuric acid.

Write equation and the mechanism for the reaction that takes place.

(5 marks)

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12. a) Define the term standard enthalpy of hydrogenation. (1 mark)

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b) The experimental values of standard enthalpy of hydrogenation of some compounds are given in the table below.

Compound	Ethene	Cyclohexene	Benzene	But a -1,3 – diene
$\triangle H_h^{\theta}$ ($KJmol^{-1}$)	-132	-120	-246	-239

Use the above data to calculate the expected enthalpies of hydrogenation of;

i) but a – 1,3 – diene (1 ½ mark)

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ii) benzene (1 ½ marks)

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c) Comment on the values obtained in (b) (2 marks)

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d) The standard enthalpy of formation of water and carbon dioxide respectively are -285.9 , -393.5KJmol^{-1} and standard enthalpy of combustion of benzene is - 3267KJmol^{-1}.

Calculate the standard enthalpy of formation of benzene. (3 marks)

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13. a) (i) Distinguish between a thermoplastic and a thermosetting plastic.(2 marks)

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(ii)Give the name of one example for each of the plastics in a(i). (1 mark)

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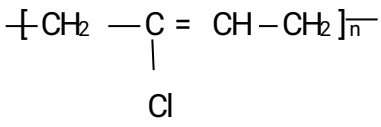
b) The structural formula of a plastic R is



Write the ;

i) structural formula and name of the monomer of R. (1 ½ marks)

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ii) name of the type of polymerization leading to the formation of R.

(0 ½ marks)

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c) The osmotic pressure of a solution containing 2.5g of R per litre of benzene is $4x10^{-4}$ atmospheres at $25^{0}C$.

i) Calculate the relative molecular mass of R. (2 marks)

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ii) Determine the value of n in the formula of R. (1 ½ marks)

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d) State one use of R. (0 ½ marks)

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14. a) (i) Write equation(s) to show how sulphur dioxide can be converted to sulphuric acid. (3 marks)

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(ii) State two sources of sulphur dioxide used in (i) (1 mark)

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b) State what would be observed and write equations for the reaction that occurs when concentrated sulphuric acid reacts with;

i) potassium bromide (2 ½ marks)

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ii) potassium iodide (2 ½ marks)

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15. State what would be observed and write equation for the reaction that would take place when

a) Aqueous solution of iodine and sodium hydroxide is mixed with propanone and the mixture warmed. (2 marks)

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b) Hydrogen peroxide is added to acidified potassium manganate (VII) solution. (2 ½ marks)

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c) Dilute sulphuric acid is added to potassium manganate (VI) solution.

(2 ½ marks)

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d) A mixture of propanal and Fehling's solution is heated. (2 marks)

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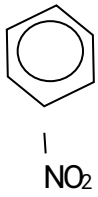
16. Complete the following equations and in each case outline the accepted mechanism leading to formation of the major organic product.

a)



$$\xrightarrow[H_3PO_4(aq)]{CH_3CH{=}CH_2}$$

(3 marks)

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b) $CH_3CH_2CH(CH_3)CH_2OH \xrightarrow[180^0C]{\text{Conc. } H_2SO_4}$ (3 ½ marks)

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c) $CH_3COBr \xrightarrow[\text{Reflux}]{\text{Conc. } NH_3}$ (2 ½ marks)

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17. a) (i) State the meaning of the term solubility product. (1 mark)

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(ii)Give one application of solubility product. (0 ½ mark)

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b) Calcium Iodate is sparingly soluble in water.

Write the;

i) equation for solubility of calcium iodate. (1 ½ mark)

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ii) expression for the solubility product , Ksp for calcium iodate.

(0 ½ mark)

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c)The solubility product , Ksp of calcium iodate is $1.69 \times 10^{-3} mol^{3} l^{-3}$ at $25^{0}C$. Calculate the;

i) molar concentration of iodate ions in a saturated solution of calcium iodate in water. (1 ½ marks)

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ii) mass of calcium iodate precipitated when 0.1M solution of calcium nitrate is added to solution in (i). (3 marks)

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d) Aqueous sodium phosphate solution was added to the mixture in c(ii).

State what would be observed and explain your answer. (1 mark)

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END